# Changing velocity

**antocciphysics**



**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Calculate acceleration and distinguish speeding up, slowing down, and turning.

**SUCCESS CRITERIA - I can...**

1. Calculate acceleration from velocity change and time.
2. Compare acceleration direction with velocity direction.
3. Explain why turning at steady speed is acceleration.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

## 01 Bring back an earlier idea

Velocity is +4 m/s for 6 s, then -3 m/s for 4 s. Find displacement and distance. Explain the reference information needed for an angle of 30°.

**Marzano check - circle:** ① ② ③

## 02 Explain a worked example

**Reference:** $a_{avg} = (v_f - v_i)/\Delta t$.
i = initial; f = final. Compare directions of velocity and acceleration to decide whether speed increases.

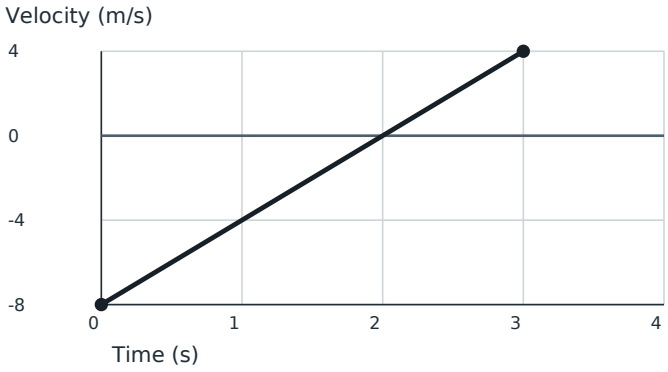
**Worked reasoning:** A cart moving east slows from 8 to 2 m/s in 3 s. Taking east positive, $a=(2-8)/3=-2$ m/s². It keeps moving east while acceleration points west.

**Your turn:** Draw initial and final velocity arrows and one acceleration arrow. Use arrow lengths to show the speed change.

**Marzano check - circle:** ① ② ③

## 03 Signed acceleration

Use the graph to calculate acceleration and explain the speed trend. State when the cart is momentarily at rest.




**Marzano check - circle:** ① ② ③

## 04 Braking intervals

A car travels at +24 m/s and slows at a constant rate to 0 m/s in 7.0 s. The positive direction is the car's original direction of travel.
(a) Calculate its acceleration.
(b) Calculate its displacement during braking.
(c) Explain the sign of each answer.

**Marzano check - circle:** ① ② ③

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Continue down each column.

### 05 Small acceleration, long time

A hypothetical constant sideways acceleration is $2\times10^{-6}$ m/s² for $10^{6}$ s. Find Δv and explain why small acceleration alone cannot justify ignoring it.

**Marzano check - circle:** 1 2 3

### 06 Choose a method

A=(5, -3) m and B=(-3, 7) m. Find R, magnitude and direction. Choose C so A+B+C=0 and sketch the route.

**Marzano check - circle:** 1 2 3

### 07 Exit check

Velocity changes from -3 to -9 m/s in 4 s, then stays -9 m/s for 4 s. Find acceleration and speed trend in each interval.

**Marzano check - circle:** 1 2 3

### 08 STRETCH: Apply it another way

A cart begins at -6 m/s and has constant acceleration +2 m/s² for 5 s. Sketch v versus t; find turning time, displacement and distance.

**Marzano check - circle:** 1 2 3